Four-Adapter Build Roadmap

Extending the RICD Collapse-Tracking Engine to Banking, Hospitals, Public Schools, and Nursing Homes

Overview

This is a proposed build plan for four new domain adapters to the Recursive Information-Container Dynamics (RICD) engine, following the same adapter-contract pattern (Part 15 of the manuscript) already validated once, concretely, by the higher-education tracker: a real implementation using only public IPEDS finance and enrollment data that reached 100% leave-one-out cross-validated accuracy across a 54-institution panel of confirmed closures and stable comparators, and flagged Hampshire College as high-risk more than three years before its actual closure. What follows is a plan to repeat that same validation exercise — real public data, a confirmed-outcome panel, and honest leave-one-out testing — in four more domains chosen because each already has some form of the same structural gap the higher-ed tracker measures: an official record that can look stable right up until the point of collapse. No schedule is attached: each adapter is a self-contained build that can proceed, or be taken much deeper, on its own.

The framing follows the higher-ed tracker's own approach: public-interest collapse tracking built on public data, aimed at surfacing structural risk earlier than official reporting shows it. All four adapters below also sit inside a real, existing commercial category worth naming directly — institutional credit-risk and early-warning analytics, already served by Moody's, S&P Global, Fitch, and KBRA for banks, hospital systems, and school district bonds, and by senior-housing lenders, long-term care insurers, and state Medicaid oversight for nursing homes. No commercial figures are included here, since anything specific would be fabricated at this stage, but the buyer category named for each adapter is real, and each has a genuine reason to want what RICD would tell it that its current tools structurally cannot — a real market existing for this category of analytics is not the same claim as confirmed demand for this specific tool, and nothing here should be read as a demand-validation exercise.

Every data source named below is a real, currently accessible public dataset, checked directly rather than assumed; where a source turned out not to be genuinely public or not to exist, it has been replaced with a real substitute or flagged openly as a gap, rather than left in as if it were solved.

Build Order

Order	Adapter	What building it involves
1	Public Schools	Adapter contract mapping → CCD/F-33 data pipeline → closure panel (built from CCD's own status flags) → calibration → validation + methods note

Order	Adapter	What building it involves
2	Banking	Adapter contract mapping → FFIEC Call Report pipeline → resolve quarterly-vs-annual timestep → FDIC failure-panel construction → calibration → validation + methods note
3	Hospitals	Adapter contract mapping → HCRIS pipeline → cross-source ID crosswalk (HCRIS / PESP / Sheps Center) → closure panel → calibration → validation + methods note
4	Nursing Homes	Adapter contract mapping → CMS cost report (Form 2540) pipeline reusing hospital-adapter tooling → PBJ staffing + ownership-data integration → closure panel verified against CMS termination records → calibration → validation + methods note

Each adapter goes through the same five phases the higher-ed tracker already validated: (1) contract mapping — assigning real inputs to the official (O_o), operational (O_p), and meta-representation channels, the resource/governance-override signal, and a confidence-type label (observed, estimated, reconstructed, or adapter-defined) for every one of them, per Part 15 of the manuscript; (2) building the data acquisition and cleaning pipeline; (3) constructing a panel of confirmed-outcome cases against matched stable comparators; (4) fitting the model and calibrating regime and hull-breach thresholds; and (5) leave-one-out (or equivalent) validation, closed out with a short methods note in the same format as the higher-ed tracker's own documentation.

The order is chosen by build risk, not by domain importance. Public schools goes first because its data — an annual federal survey combining finance and enrollment, with closures already flagged in the source data — is the closest structural match to IPEDS, so it is the fastest way to prove the adapter-contract pattern generalizes at all beyond the domain it was built for. Banking goes second specifically because it is where the one real engineering question in this whole plan surfaces: Call Reports are filed quarterly, a finer time resolution than the annual cadence the current model was built around, and this adapter is where that gets resolved — either by resampling to an annual window or by extending the model's timestep. Hospitals goes third because it is the most assembly-heavy of the four: no single source covers official financials, operational staffing, ownership shocks, and confirmed closures, so three independently-run datasets have to be cross-walked onto the same institutions before any modeling starts, and that is real, un-skippable labor. Nursing homes goes last mainly because it should benefit directly from going right after hospitals: CMS runs the cost-report infrastructure for both domains (Form 2540 for nursing homes is the direct structural sibling of the hospital adapter's Form 2552), so the data-cleaning conventions, confidence-typing choices, and validation harness built for hospitals should carry over with real savings rather than starting from a blank page.

Public Schools Adapter

Official layer (O_o): NCES Common Core of Data (CCD) district and school directory and membership/enrollment files, plus the Census Bureau's Annual Survey of School System

Finances (F-33) for revenue and expenditure by source — the closest existing match to IPEDS finance data of any domain here, run by the same combination of federal statistical agencies.

Operational layer (O_p): Pupil-teacher ratio trends from CCD's staff files, and chronic absenteeism / discipline data from EDFacts and the Civil Rights Data Collection. The CRDC is collected on a slower, biennial cadence than the annual finance data, which the adapter has to handle explicitly rather than silently interpolate.

Meta / governance-override channel: State-level fiscal-distress or "financial watch" designations and state takeover/receivership actions. These are real and public but published per-state with no single federal aggregator, so this channel is built as an adapter-defined, hand-compiled signal rather than an automated feed.

Confirmed-outcome panel: CCD's own directory files already flag schools and districts as closed or consolidated historically, which gives a built-in confirmed-closure list without needing an external tracker — genuinely the simplest panel-construction task of the four adapters.

Commercial application: The same market Moody's, S&P Global, and Fitch already serve rating school district general-obligation bonds for municipal bond investors and state oversight offices. Several states already run their own fiscal early-warning systems for districts; this adapter is a candidate to compete with or supply those systems rather than invent a buyer from nothing.

Main build risk: Low. The principal work is reconciling the different update cadences across CCD, F-33, and CRDC, not sourcing the data itself.

Banking Adapter

Official layer (O_o): FFIEC Call Reports, Schedule RC-R (Common Equity Tier 1, Tier 1 capital, leverage ratio, liquidity coverage ratio), pulled directly through the FFIEC Central Data Repository's public bulk data distribution — confirmed live and downloadable, not a proprietary feed.

Operational layer (O_p): Uninsured and wholesale deposit share (Schedule RC-O) and the held-to-maturity/available-for-sale unrealized loss position relative to portfolio duration (Schedule RC-B) — both computable directly from the same public Call Report filings. This replaces the non-public interbank repo flow and internal audit data an earlier, AI-drafted version of this plan had proposed, which turned out not to be accessible to an independent researcher at all.

Meta / governance-override channel: Public enforcement actions: the FDIC's own Enforcement Decisions and Orders database, the OCC's enforcement actions search, and Federal Reserve enforcement press releases — all public and searchable, serving the same role the accreditor show-cause order plays in the higher-ed tracker.

Confirmed-outcome panel: The FDIC's BankFind Suite failed-bank list, public back to 1934 — almost certainly the largest and cleanest confirmed-failure panel available across any of the four domains, with matched stable comparators drawn from banks of similar size and charter type that never failed.

Commercial application: Bank counterparty and correspondent-banking risk scoring, the same market already served by Moody's, S&P Global Market Intelligence, and KBRA, sold to bank treasury departments managing exposure to other institutions and to a bank's own board risk committee — not to parties betting against the bank's survival.

Main build risk: Medium, and structural rather than a data-sourcing problem: Call Reports are quarterly, while the existing model's state-space formulation was built around IPEDS' annual cadence. Deciding how to handle that mismatch is real, undesigned engineering work, which is exactly why this adapter is scheduled early enough to have room to get it right.

Hospitals Adapter

Official layer (O_o): CMS's Healthcare Cost Report Information System (HCRIS) hospital cost reports, and Form 990 Schedule H community-benefit filings for nonprofit hospitals — the direct structural analog to IPEDS finance data in this domain.

Operational layer (O_p): Staffing intensity (FTEs per adjusted discharge or per bed) and Medicaid/uncompensated-care share of payer mix, both derivable from HCRIS worksheets, supplemented where available by CMS Care Compare staffing data.

Meta / governance-override channel: Ownership-change events from the Private Equity Stakeholder Project's Private Equity Hospital Tracker — confirmed as an actively maintained, 2026-updated public dataset built for exactly this kind of tracking, not something that has to be assembled from scratch — plus state-level financial-distress or closure-warning designations where states publish them.

Confirmed-outcome panel: UNC's Sheps Center Rural Hospital Closures database as the primary confirmed-closure list (actively maintained, includes a closure map and periodic state-of-rural-hospitals reporting), supplemented by the American Hospital Association's broader annual closure counts for a larger, not rural-only, comparator pool.

Commercial application: Nonprofit and health-system credit-risk analytics, already a real market served by Moody's and Fitch's healthcare ratings groups, plus a direct-to-operator self-monitoring product sold to hospital system CFOs and boards wanting independent early warning of the kind of PE-driven structural decline conventional accounting can mask.

Main build risk: The highest data-assembly burden of the four. HCRIS, the PESP tracker, and the Sheps Center closures list each use different institutional identifiers, so a real crosswalk across CMS certification numbers, PESP's own IDs, and Sheps Center's facility names has to be built before modeling starts. This is the reason this adapter gets the most calendar time in the plan.

Nursing Homes / Long-Term Care Adapter

Official layer (O_o): CMS Nursing Home Compare / Care Compare five-star ratings and self-reported facility data, plus nursing home Medicare cost reports (Form 2540) — the direct structural sibling of the hospital adapter's own HCRIS cost reports, filed to the same agency.

Operational layer (O_p): CMS Payroll-Based Journal (PBJ) staffing data — payroll-verified, quarterly, facility-level nurse-hours, live since 2016 — a stronger operational proxy than self-reported staffing because it is checked against payroll records rather than taken on the facility's word.

Meta / governance-override channel: CMS's own nursing home ownership data, public since 2023 and including private-equity ownership flags, as the ownership-shock signal, plus Special Focus Facility program designations and civil monetary penalties from CMS's inspection and enforcement records. A 2023 GAO report found real gaps and discrepancies in that ownership dataset, so it is usable but not clean, the same caveat that applies to the hospital adapter's PE-ownership data.

Confirmed-outcome panel: A peer-reviewed study (Health Affairs Scholar) found 1,459 confirmed nursing home closures between 2011 and 2021, 8.92% of all U.S. facilities over that decade, with a documented year-by-year rate — a panel two orders of magnitude larger than anything available for open-source software, to be built by cross-referencing that published count directly against CMS's own Provider of Services termination records rather than relying on the published estimate alone. One industry fact sheet cites a much larger cumulative figure, 44,459 closures since 2015, which looks inconsistent with the peer-reviewed count and should not be used until it's clear what it actually measures — possibly bed closures rather than facility closures.

Commercial application: Senior housing REITs and lenders (Ventas, Welltower, HUD Section 232 loan underwriters), long-term care insurers, and state Medicaid oversight agencies all carry direct financial exposure to the structural decline this adapter is built to catch. A facility operator is also a plausible buyer, purchasing the score as an internal early-warning tool to catch problems before a CMS survey does — the same “customers who want to be monitored” framing that made this domain preferable to doing the same thing for open-source software.

Main build risk: Mostly inherited and reduced, not new: crosswalking CMS's own cost report, PBJ, ownership, and Care Compare datasets should be more tractable than the hospital adapter's cross-organization crosswalk, since all four nursing-home sources are published by the same agency. The real open risk is the ownership dataset's GAO-flagged quality gaps, which limits how much weight that signal alone should carry until checked against the closure panel.

Honest Risks

This is the first attempt to generalize RICD beyond the domain it was originally built and validated on, and three risks are worth naming directly rather than leaving implicit. First, the banking adapter's quarterly-versus-annual timestep question is genuinely undesigned right now; if it turns out to require real changes to the underlying state-space model rather than a resampling fix, that adapter becomes a substantially larger build than the other three. Second, the hospital adapter's three-source crosswalk (HCRIS, the PESP tracker, and Sheps Center closures) is the kind of data-matching work that reliably takes longer than initial estimates in almost any domain. Third, the nursing home adapter's confirmed-outcome panel currently rests on a published estimate (1,459 closures, 2011–2021) rather than a raw count pulled directly from CMS's own termination records, and its ownership-data channel has documented, GAO-flagged quality gaps

— both need to be checked against the primary source before the adapter's validation numbers go into anything final, not assumed to hold at the scale currently cited.